# Redesigning a Digital Interaction for Better Usability

## Sri Lanka Railways Website

INTE 31233 – Human Computer Interaction

Group 14

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# ABSTRACTION

Public transportation websites play a critical role in providing accurate and timely information to passengers. The Sri Lanka Railways website serves as the official digital platform that allows users to check train schedules, make seat reservations, and view important railway announcements. However, several usability and accessibility challenges reduce the effectiveness of the system and make navigation difficult for many users. This study analyzes the usability problems present in the Sri Lanka Railways website and proposes an improved interface based on Human–Computer Interaction (HCI) principles. The research involved identifying usability issues, developing user personas, analyzing user tasks, and proposing redesigned wireframes to improve navigation and readability. The redesigned interface aims to enhance user efficiency, reduce confusion, and improve accessibility for users with different levels of technical knowledge. The results demonstrate that applying structured layout design, clear navigation elements, and improved accessibility features significantly improves overall user experience. This study highlights the importance of usability-focused redesign in public digital services.

# I. INTRODUCTION

In modern society, digital platforms are widely used to provide public services such as transportation, banking, and communication. Transportation websites play a particularly important role because they provide passengers with real-time information regarding travel schedules, ticket availability, and service updates. A well-designed transportation website should allow users to access information quickly, easily, and accurately.

The Sri Lanka Railways website is an official government platform used by thousands of passengers daily. The website provides several services, including train timetable viewing, seat reservation, fare checking, and access to railway announcements. Although the system provides important functionality, users often experience usability challenges that affect efficiency and satisfaction. For example, users may struggle to locate important information such as train schedules or reservation links due to unclear navigation structure and crowded page layouts.

Human Computer Interaction principles emphasize designing systems that are user-centered, efficient, and accessible. Applying these principles helps reduce user errors, improve system learnability, and increase satisfaction. Therefore, this project focuses on analyzing usability problems in the Sri Lanka Railways website and redesigning the interface to improve overall usability and accessibility.

The main objectives of this study include identifying usability issues, understanding user behavior through research, proposing improved interface designs, and evaluating usability improvements using structured testing methods.

## II. PROBLEM IDENTIFICATION

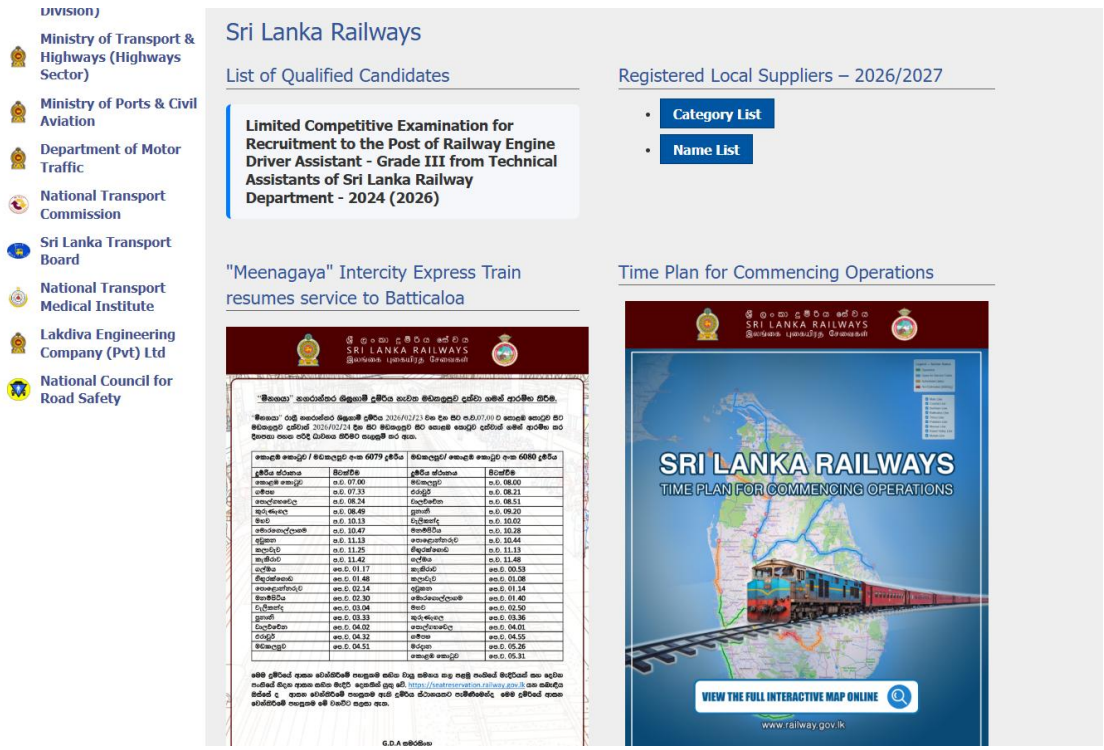
The selected system for analysis in this project is the official Sri Lanka Railways website. Initial observation and interaction with the website revealed several usability problems that negatively affect user experience. These problems relate to navigation complexity, readability issues, and limited accessibility features.

## A. Poor Information Structure

One of the major usability issues observed in the Sri Lanka Railways website is the poor organization of information. Important content such as train schedules, railway notices, and service announcements are scattered across different sections of the website. Users often need to scroll extensively and explore multiple pages before finding the information they require.

This issue increases the time required to complete tasks and creates confusion among users. For example, users searching for train schedules may need to examine multiple menus before identifying the correct section. A well-structured information hierarchy is essential to ensure that users can locate information quickly without unnecessary effort.

Poor information structure also affects first-time users who are unfamiliar with the system layout. Without clear grouping of related information, users may struggle to understand where to begin their search.



## B. Difficult Navigation

Navigation complexity is another significant usability issue identified during system evaluation. The navigation menu contains numerous options without clear categorization, making it difficult for users to identify commonly used services.

For example, essential features such as train timetable search and seat reservation are not prominently displayed on the homepage. Users must navigate through several links to locate these functions. This increases cognitive load and reduces efficiency, especially for users who require quick access to information.

Effective navigation design should prioritize frequently used features and provide clear pathways to complete tasks. Without proper navigation structure, users may feel frustrated and abandon the system.

## C. Readability Issues

Readability problems were also observed in several sections of the website. Many announcements and notices are displayed as scanned images instead of formatted text. This makes it difficult for users to read information quickly, especially when using mobile devices with smaller screens.

Long paragraphs of text without proper spacing further reduce readability. Users often find it difficult to scan information and identify important details. In addition, small font sizes contribute to visual strain, particularly for elderly users.

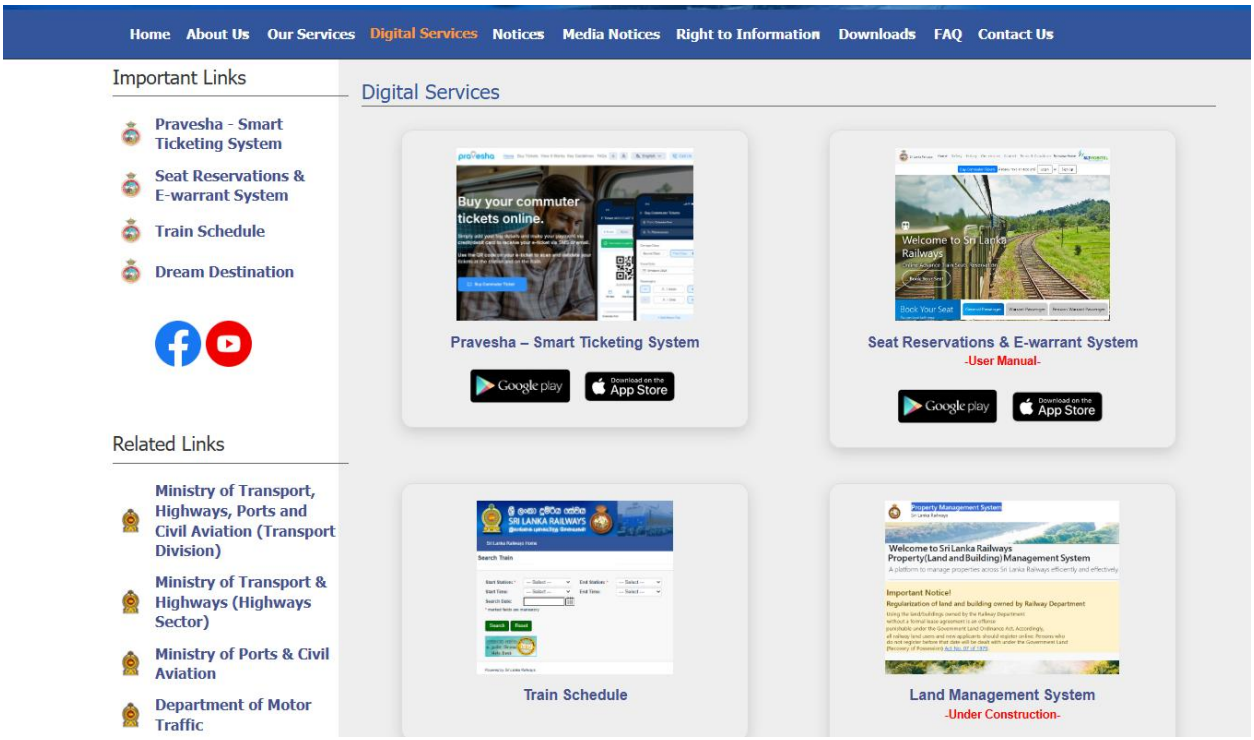
Improving readability requires using structured text formats, appropriate font sizes, and proper spacing between content sections.

## D. Limited Accessibility

Accessibility limitations were identified as another major concern. The website lacks accessibility features that support users with visual impairments or limited technical skills. For example, image-based content cannot be interpreted by screen readers, making it inaccessible for visually impaired users.

Low contrast between text and background also reduces readability. Accessibility standards recommend using sufficient color contrast and scalable text sizes to improve visibility. Without

these features, users with disabilities may face significant challenges when interacting with the system.




## E. Cluttered and Inconsistent Design

The homepage contains multiple banners, links, and text blocks arranged without consistent alignment. This cluttered layout creates visual overload and makes it difficult for users to focus on important elements.

Inconsistent use of colors, fonts, and buttons across different pages further increases confusion. Users expect similar functions to appear in consistent locations throughout the system. When design patterns change unexpectedly, users require additional effort to understand the interface.

A simplified and consistent layout is necessary to improve usability and reduce cognitive load.

# III. USER RESEARCH

User research plays a critical role in understanding how different users interact with the system. In this study, user personas were developed to represent typical railway passengers with varying needs and technical abilities. Personas help designers identify user goals, frustrations, and expectations, enabling more effective design decisions.

## A. Persona 1 – University Student

Shenal Perera is a 22 year old university student who frequently travels between Galle and Colombo. He has moderate technical skills and regularly uses digital platforms to access travel information.

Shenal typically visits the railway website to check train schedules, verify ticket prices, and locate reservation links. His primary goal is to find travel information quickly without unnecessary delays. However, he often has trouble locating timetable sections due to unclear navigation structure.

One of Shenal's main frustrations is the time required to search through multiple pages before finding relevant information. In addition, the website layout sometimes appears confusing on mobile devices, making it difficult to complete tasks efficiently.

From a design perspective, Shenal's needs suggest the importance of providing a prominent timetable search feature on the homepage. Large buttons and simplified navigation menus can significantly improve efficiency for users with similar characteristics.

## B. Persona 2 - Office Worker Traveler

Nadeesha Fernando is a 32-year-old office executive who relies on daily train transportation to travel to work. She has strong technical skills and uses her mobile device extensively throughout the day.

Nadeesha values efficiency and prefers systems that allow quick access to information. She often uses the railway website during busy hours to verify train schedules and reservation availability. However, she frequently encounters delays due to slow navigation and unclear menu structure.

Her main challenge involves locating reservation links quickly without navigating through multiple pages. She also expects mobile-friendly layouts that support efficient interaction during travel.

Design improvements for users like Nadeesha include streamlined navigation pathways, responsive layouts, and reduced page loading time.

## C. Persona 3 - Elderly Passenger

Sunil Jayasinghe is a 60-year-old retired passenger who occasionally travels using railway services. He has limited technical knowledge and prefers simple interfaces with clear instructions.

Sunil often experiences difficulty reading small text and understanding complex navigation structures. He requires large fonts, clear labels, and minimal distractions within the interface.

For users with similar characteristics, accessibility improvements such as increased font size, high contrast colors, and simplified layout are essential.

# IV. DESIGN AND PROTOTYPING

After identifying usability problems and understanding user needs through research, the next phase of the project involved designing improved interface solutions. The design phase focused on applying Human Computer Interaction (HCI) principles to improve usability, accessibility, and overall user experience.

The redesign process began by identifying the most common tasks performed by users on the Sri Lanka Railways website. Based on these tasks, alternative layouts were developed using simplified structures, improved navigation elements, and clearer visual hierarchy. The design decisions were guided by usability principles such as predictability, consistency, and familiarity.

## A. Task Analysis

Task analysis was conducted to understand how users currently interact with the railway system and to identify inefficiencies in task completion. The selected task for analysis was searching for train schedules, as this is one of the most frequently performed activities on the website.

In the existing system, users begin by opening the homepage and scrolling through multiple sections to locate the train schedule link. After finding the link, users are redirected to another page where they enter departure and destination stations. This process requires multiple steps and creates confusion, especially for first-time users.

Several problems were observed during this task,

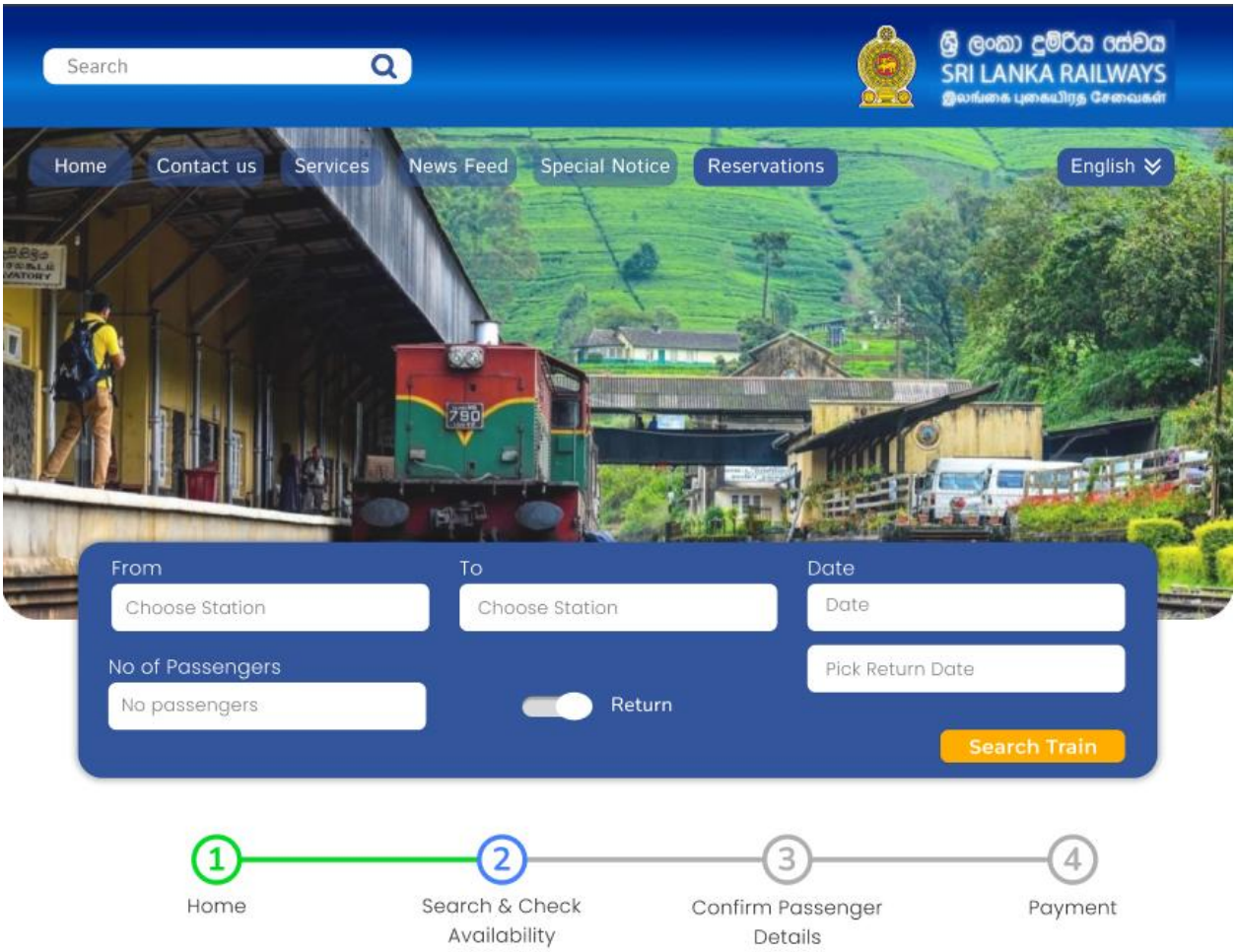
- Users required excessive scrolling to locate schedule links
- Navigation paths were unclear
- Some links redirected to external pages
- Users were unsure whether actions were successful

To improve efficiency, a redesigned task flow was proposed. In the new design, the timetable search feature is placed prominently on the homepage. Users can immediately enter departure and destination details without navigating through multiple pages.

This improvement reduces task completion time and improves user confidence.

## B. Design Goals

The primary objective of the redesign was to create a user-friendly interface that improves accessibility and reduces user frustration. Several design goals were established to guide the redesign process. The first goal was to improve navigation clarity by placing frequently used features such as timetable search and seat reservation on the homepage. This reduces the number of steps required to complete tasks. The second goal was to enhance readability by increasing font size, improving contrast, and organizing content into structured sections. This improvement ensures that users can read information comfortably across different devices. Another important goal was to improve accessibility by minimizing clutter and simplifying interface elements. This ensures that users with limited technical skills can interact with the system easily. Finally, consistency across pages was prioritized to ensure that users encounter familiar layouts throughout the system.



## C. Application of HCI Principles

Several Human Computer Interaction principles were applied during the redesign process to ensure usability improvements.

### *Predictability*

Predictability refers to the user's ability to anticipate system behavior. In the redesigned interface, clear labels and icons were added to navigation buttons to ensure that users understand what will happen when they select an option. For example, selecting the Train Schedule button immediately displays the timetable search form.

This improvement ensures that users can interact with the system confidently without confusion.

### *Synthesizability*

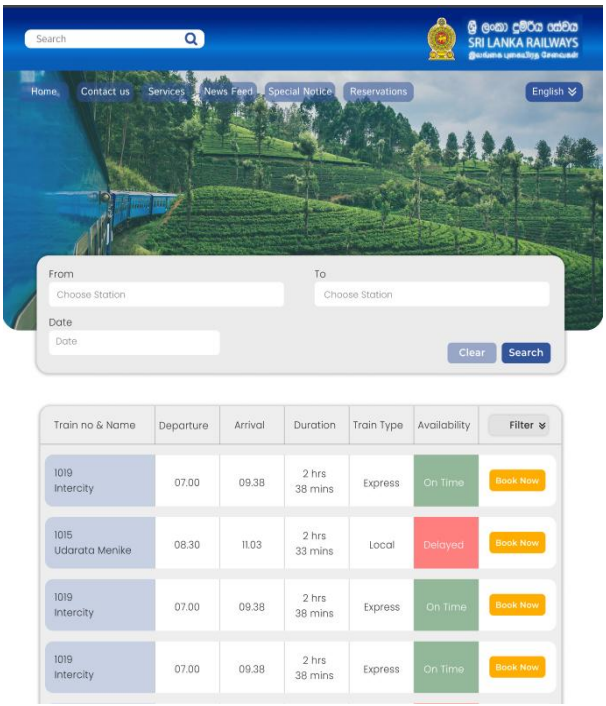
Synthesizability refers to the user's ability to understand the outcome of their actions. In the redesigned interface, confirmation messages were added to indicate successful actions such as completing a search or submitting reservation details.

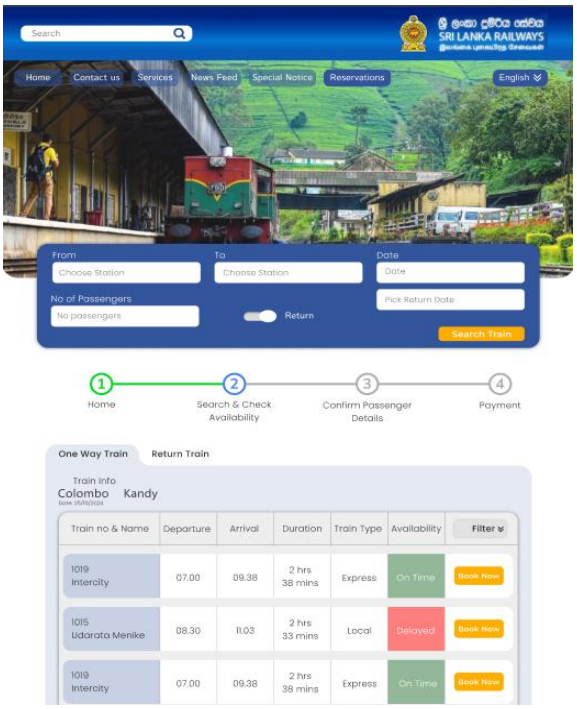
Loading indicators were also included to inform users when the system is processing requests. These features provide clear feedback and improve system transparency.

### *Generalizability*

Generalizability allows users to apply previously learned interactions to new situations. In the redesigned system, consistent layouts were used across different pages, allowing users to transfer their knowledge from one section to another.

For example, timetable search fields follow the same layout structure as reservation forms, enabling users to perform tasks efficiently without relearning navigation patterns.

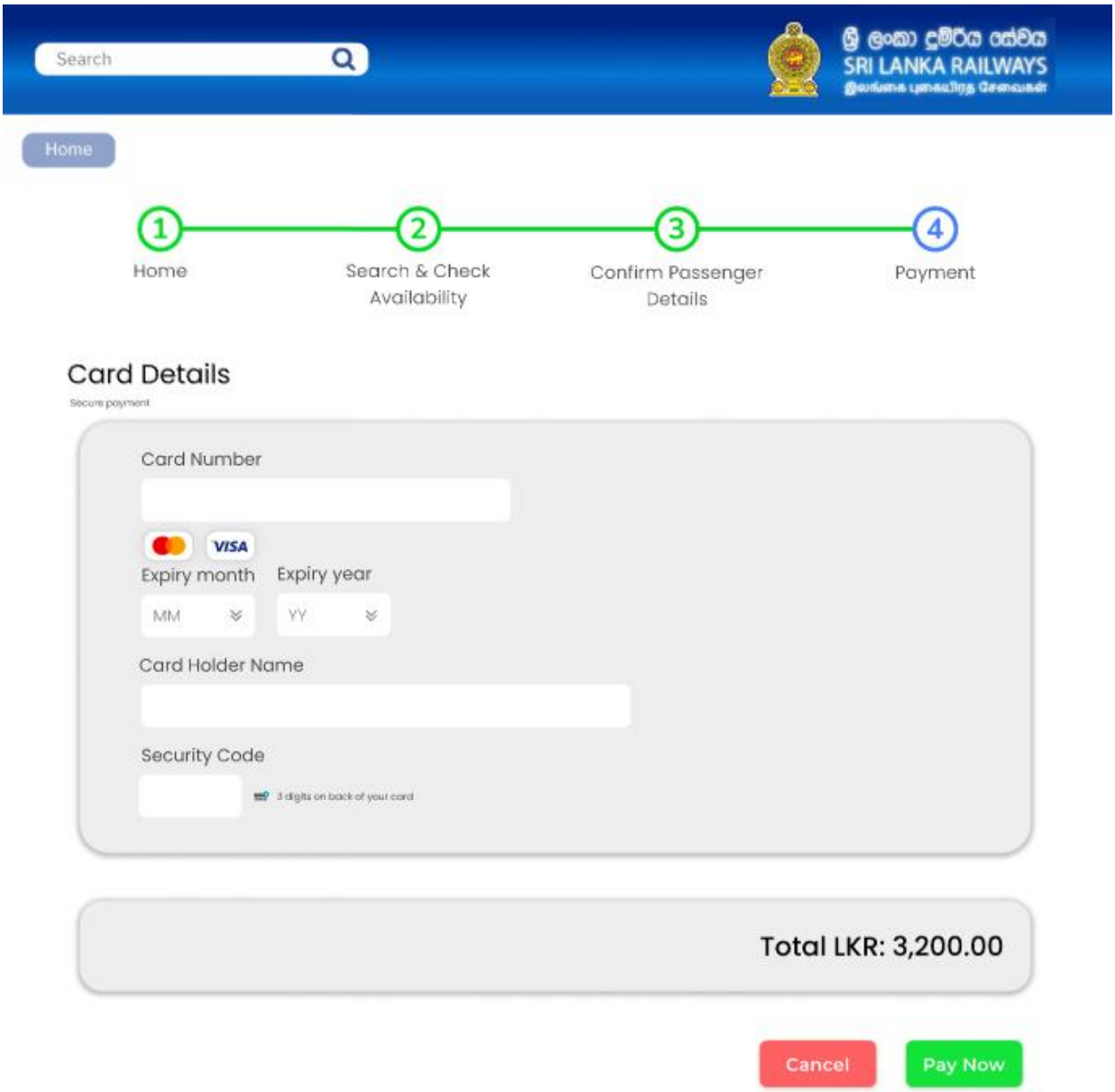


### *Familiarity*

Familiarity ensures that users recognize interface elements based on prior experience. The redesigned interface uses common button styles such as Search, Book Now, Pay Now and Cancel, which are widely recognized across digital systems.

Using familiar icons and labels reduces learning time and improves usability.



*Consistency*

Consistency ensures that visual elements remain uniform across the interface. The redesigned layout uses standardized fonts, colors, and button styles to create a predictable user experience.

Consistent placement of navigation menus ensures that users can locate functions quickly across different pages.

### D. Prototype Development

Based on the design goals and usability principles, wireframes were developed to represent the redesigned interface. The prototype focused primarily on improving the homepage layout and navigation structure.

The redesigned homepage includes large navigation buttons for major services such as Train Schedule, Seat Reservation, and Service Notices. A centralized search section allows users to quickly enter travel details.

Content sections were organized into structured blocks with clear headings, reducing clutter and improving readability. The layout was also optimized for mobile devices to ensure responsive interaction across different screen sizes.

The prototype represents an improved visual hierarchy that guides user attention toward important elements.

## V. USABILITY TESTING AND EVALUATION

Usability testing was conducted to evaluate the effectiveness of the redesigned interface. Testing focused on measuring user performance and identifying remaining usability challenges.

### A. Testing Method

The Think-Aloud Protocol was used as the primary testing method. Participants were asked to perform specific tasks while describing their thoughts and actions. This method helps identify confusion points and usability issues.

The selected tasks included,

• Searching for train schedules
• Locating reservation links
• Viewing service announcements

Participants completed tasks using both the existing and redesigned interface.

## B. Test Participants

A total of 12 participants were selected for usability testing. These participants represented different user groups, including students, office workers, and elderly passengers.

The selection ensured diversity in technical skill levels and travel frequency, allowing realistic evaluation of usability improvements.

## C. Survey Results

After completing usability tasks, participants were asked to provide feedback through a structured questionnaire. The survey included questions related to navigation clarity, readability, and overall usability.

The results indicated that most users found the redesigned interface easier to use than the existing system. Participants reported reduced confusion and faster task completion times.

Users particularly appreciated the simplified navigation layout and improved readability.

## D. Task Completion Time Analysis

Task completion time was measured to evaluate efficiency improvements.

Before redesigning, users required significant time to locate schedule information due to navigation complexity. After redesign, the simplified layout reduced the number of required steps, resulting in faster task completion.

For example, the average time required to locate train schedules decreased significantly after implementing the redesigned interface. Similarly, reservation link access time improved due to clearer navigation pathways.

These results demonstrate measurable improvements in usability performance.

## E. Improvements After Testing

Based on feedback from participants, several additional improvements were made to the prototype.

Font sizes were increased to improve readability, particularly for elderly users. Icons were added next to navigation buttons to improve clarity. Menu structure was simplified further to minimize confusion.

These refinements improved usability and ensured that the redesigned system met user expectations.

## VI. DISCUSSION

The redesigned railway website demonstrates significant improvements in usability and accessibility compared to the existing interface. By applying structured layout design and navigation improvements, the system reduces cognitive load and improves efficiency.

Users were able to complete tasks more quickly due to the simplified navigation structure. Improved readability features enhanced user comfort, particularly for users accessing the website through mobile devices.

Accessibility improvements also ensured that users with limited technical knowledge could interact with the system more easily. Overall, the redesigned interface supports better user experience and improved task performance.

## VII. CHALLENGES FACED

Several challenges were encountered during the project development process.

One major challenge involved collecting sufficient user feedback within limited time constraints. Coordinating usability testing sessions required careful planning to ensure participant availability.

Another challenge involved balancing simplicity with functionality. While simplifying navigation was necessary, it was important to retain essential features within the interface.

Ensuring compatibility across different devices also presented technical challenges. Responsive layout design required additional effort to maintain usability across various screen sizes.

Despite these challenges, the project successfully implemented meaningful usability improvements.

## VIII. FUTURE IMPROVEMENTS

Although the redesigned interface improves usability, several additional features can further enhance system performance.

Future improvements may include implementing multilingual support to accommodate users from different linguistic backgrounds. Voice-based navigation can also improve accessibility for visually impaired users.

Real-time train tracking features can provide passengers with accurate information about delays and arrival times. Additionally, developing a dedicated mobile application may improve convenience for frequent travelers.

Continuous usability evaluation will ensure that the system evolves according to user needs.

## IX. CONCLUSION

This project analyzed usability challenges in the Sri Lanka Railways website and proposed a redesigned interface using Human–Computer Interaction principles. The study identified several key issues related to navigation complexity, readability limitations, and accessibility barriers.

User research and usability testing revealed that simplifying navigation structure and improving visual hierarchy significantly enhanced task efficiency. The redesigned interface reduced confusion and improved user satisfaction across different user groups.

Overall, the project demonstrates the importance of user-centered design in improving public digital services. Applying usability principles not only improves efficiency but also enhances accessibility and user confidence.

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Figma Prototype Link

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